

10 CFR 50.73

RA23-023

June 16, 2023

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

LaSalle County Station, 2
Renewed Facility Operating License No. NPF-18
NRC Docket No. 50-374

Subject: Licensee Event Report 2022-003-02, Manual Scram due to Isophase Bus
Duct Fire Followed by 2A RPS Normal Power Supply Trip

In accordance with 10 CFR 50.73(a)(2)(iv)(A), Constellation Energy Generation, LLC
(CEG) is submitting Licensee Event Report (LER) Number 2022-003-02 for LaSalle
County Station, Unit 2.

There are no regulatory commitments in this letter. Should you have any questions
concerning this report, please contact Mr. Dan Mearhoff, Regulatory Assurance
Manager, at (815) 415-2800.

Respectfully,



Christopher Smith
Plant Manager
LaSalle County Station

Enclosure: Licensee Event Report

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – LaSalle County Station



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk alt: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

LaSalle County Station, Unit 2

2. Docket Number

05000 - 374

3. Page

1 OF 3

4. Title

Manual Scram due to Isophase Bus Duct Fire followed by 2A RPS Normal Power Supply Trip

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
09	26	2022	2022	003	02	06	16	2023	NA	NA
									Facility Name	Docket Number
									NA	NA

9. Operating Mode	10. Power Level
1	100 percent

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
CJ Smith, Operations Director	(815) 415-2200

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	EF	RHE	O026	Y	B	EL	IPBU	D312	Y

14. Supplemental Report Expected

☒ No☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month Day Year

6. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

At 0238 CDT on September 26, 2022 with Unit 2 in Mode 1 at 100 percent power, a manual scram was initiated due to a reported fire on the isophase bus duct. The scram was uncomplicated with all systems responding normally with the exception of a loss of 2A reactor protection system (RPS) normal power supply. The 2A RPS normal power supply output breaker tripped on over voltage. Operations successfully transferred 2A RPS power to the alternate power supply. The fire was reported extinguished at 0240 CDT on September 26, 2022.

The cause of the 2A RPS loss of normal power was a degraded voltage adjust potentiometer (POT) on the output breaker. The cause of the isophase bus duct (IPBD) fire was incorrect joint compound having been supplied and subsequently used during planned maintenance, resulting in high resistances at the shunt bolted connections.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
LaSalle County Station, Unit 2	05000- 374	2022	- 003	- 02

NARRATIVE**Plant and System Identification**

LaSalle County Station Unit 2 is a General Electric Boiling Water Reactor with 3546 Megawatts Thermal Rated Core Power. The affected systems were the Main Power system (MP) and the Reactor Protection System (RPS).

The MP system provides safe and reliable distribution of electrical power from the Main Generator to the off-site transmission network.

RPS initiates a rapid insertion of all the control rods (scram) when monitored variables exceed their pre-established limits, neutron flux instrumentation becomes inoperable, or a manual scram signal is inserted by the operator. In addition, the setpoints, power sources, and controls and instrumentation are arranged in such a manner as to preclude spurious scrams.

Condition Prior to Event

Unit(s): 2	Date: September 26, 2022	Time: 0238 CDT
Reactor Mode(s): 1	Mode(s) Name: Power Operation	Power Level: 100%

DescriptionFire and manual scram

Electrical Maintenance personnel identified smoke coming from the general area in the Unit 2 Turbine Building that contains the Unit 2 isolated phase bus duct and notified the Main Control Room (MCR). The Field Supervisor was dispatched and at 0230 CDT they notified the MCR that there was visible sparks, smoke and indications of a fire from the bus duct enclosure. Operations dispatched the fire brigade and entered abnormal operations procedures for a fire. The unit was manually scrammed at 0238 CDT. The brigade was able to successfully gain control of the fire using dry chem and the fire was reported extinguished at 0240 CDT.

2A RPS Breaker trip

Following the Manual Scram, the 2A RPS Breaker normal feed tripped on overvoltage. All Primary Containment isolations occurred as expected following the loss of A RPS power. Operations entered abnormal operating procedures, swapped 2A RPS power to the alternate feed and restored containment isolations.

Cause

The cause of the isophase bus duct (IPBD) fire was incorrect joint compound having been supplied and subsequently used during the planned maintenance performed during the previous Unit 2 refueling outage in 2021. This resulted in high shunt bolted connection resistance causing overheating in the bolted connections. The overheating resulted in a failure of the IPBD support structure hardware and ignition of the retractable cover neoprene gasket.

The cause of the 2A RPS Breaker trip was a degraded voltage adjust potentiometer.

Reportability and Safety AnalysisFire and manual scram

The fire was extinguished in less than 15 minutes of identification and the reactor was safely shutdown. Offsite power sources and station emergency power sources were available at all times throughout the event. Emergency response personnel acted promptly to extinguish the fire and there was no personnel injury during this event or damage to plant property from the fire other than the bus duct itself.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503, e-mail: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
LaSalle County Station, Unit 2	05000- 374	2022	- 003	- 02

NARRATIVE

RPS actuation is reportable under 10 CFR 50.72(b)(2)(iv)(B) as an event or condition that results in the actuation of the RPS when the reactor is critical. An ENS report was made to the NRC at 0639 on September 26, 2022 (EN# 56120).

2A RPS Breaker trip

The RPS is divided into two trip systems that are physically and electrically independent. The design of this system is such that the loss of power to one of these trip systems neither prevents nor causes a reactor scram. Normal power to RPS buses A and B is supplied by two motor-generator (MG) sets. Alternate power for either RPS bus is from the Alternate Instrument and RPS Bus Transformer. The loss of an RPS bus and associated RPS actuation was considered an invalid actuation because it was not the result of a valid signal and not an intentional manual action. The RPS bus loss condition resulted in general containment isolation signals that affected containment isolation valves in more than one system or multiple main steam isolation valves while the unit was critical. Therefore, the RPS bus loss condition is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A).

Safety System Functional Failure Review

The system equipment responses did not result in a safety system functional failure (SSFF) as defined in accordance with NEI 99-02, "Regulatory Assessment Performance Indicator Guideline." The RPS logic safety function was satisfied by its actuation for designed inputs.

Corrective Actions

Corrective actions taken in response to the conditions were:

- Unit 2 Manually scrambled.
- Fire brigade dispatched and fire promptly extinguished.
- Swap 2A RPS to alternate power supply and restored containment isolations
- Degraded 2A RPS voltage potentiometer replaced and normal power to 2A RPS restored.
- Damaged connections in the isophase bus duct restored and Unit 2 restarted.
- Incorrect joint compound removed from the Unit 2 and bolted connections were replaced with welded flexible connections.
- Increased thermographic monitoring of the Unit 1 IPBD.
- Incorrect joint compound removed from the supply system.

Previous Occurrences

LER 374-2021-002-01, Reactor Protection System Half Scram due to Motor Generator Set Output Breaker Trip. On May 31, 2021, the 2A RPS normal power supply output breaker tripped on overvoltage. The cause of this event was a degraded POT.

Component Failure Data

Device: 2A RPS Logic MG Set Output Breaker – Voltage Adjustment Potentiometer

Component Type: Rheostat / Potentiometer [RHE]

Manufacturer: Ohmite [O026]

Part: RHS500

Device: Bus, Iso Phase, Main Power Transformer

Component Type: Bus, isolated phase [GBU]

Manufacturer: Delta Unibus Corp [D312]

Part: 706232